# Beyond Reduced-Order Control: Learning the Coupled Nine-DOF Nonlinear Dynamics of a Wave Energy Converter

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Honours research with WaveX

THE UNIVERSITY OF WESTERN AUSTRALIA · WaveX · NEURAL INFORMATION PROCESSING SYSTEMS

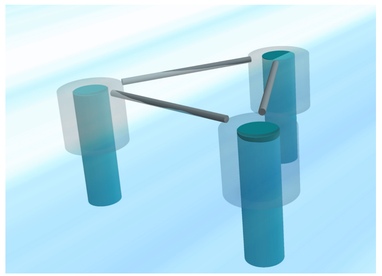
**CSSE 50th anniversary**

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## THE CONTROL CHALLENGE

### How hard should we resist the next wave?

WaveX's D-Spar converts relative motion into power. Selecting PTO damping changes both the resistance and the resulting motion.

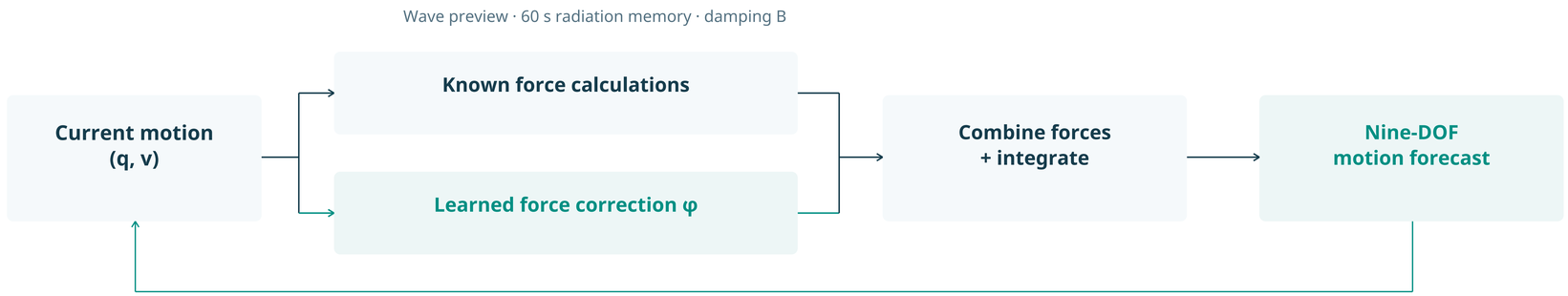


**Three linked D-Spars. Nine coupled degrees of freedom.**

Goal: retain nonlinear dynamics while making predictive control fast enough to use.

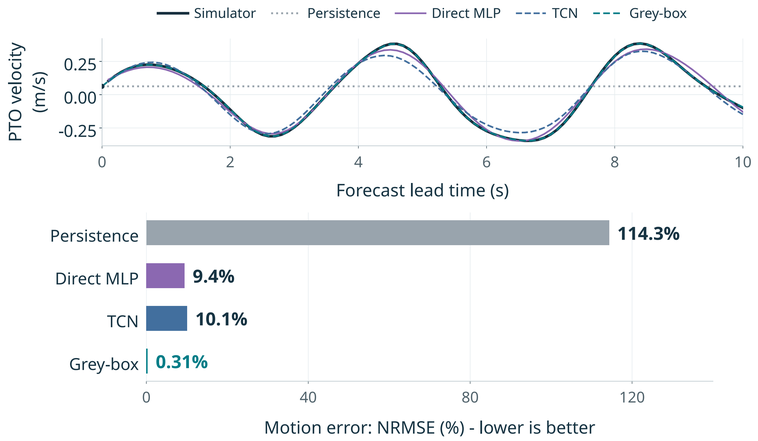
## LEARNING THE NONLINEAR DYNAMICS

Learn nonlinear drag and geometry-dependent buoyancy/gravity corrections; retain known mechanics and radiation memory.




## PREDICTING MOTION

Persistence and learned models on the same 10 s forecast outcomes.

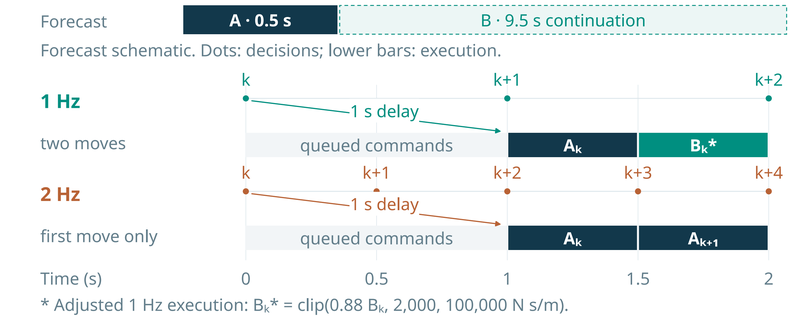



Earlier frozen surrogate benchmark: 2 waves × 8 dampings.
Same scored outcomes; models retain their native inputs.

## NONLINEAR PREDICTIVE CONTROL

POD: 4 anchor energies per continuation.
Reconstruct 41 suffix choices for each first move.

**8 first moves × 41 suffixes = 328 plans**



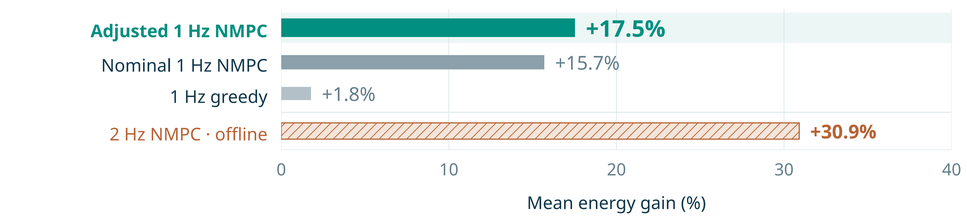

* Adjusted 1 Hz execution: $B_k^* = \mathrm{clip}(0.88\,B_k,\ 2{,}000,\ 100{,}000\ \mathrm{N\,s/m})$.

Adjusted 1 Hz: second executed command × 0.88, within damping limits. Both rates retain a 1 s application delay.

## RESULTS IN NUMERICAL SIMULATION

**+17.5%**

**mean mechanical energy adjusted 1 Hz NMPC**

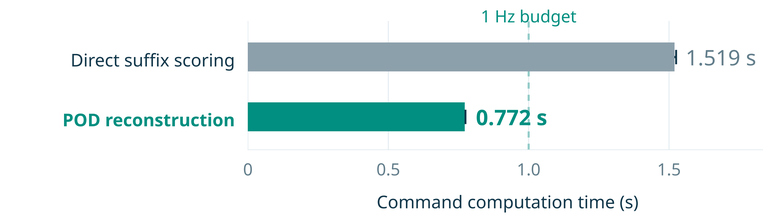



2 Hz is offline: 0.77 s median computation exceeds its 0.5 s budget.

Mean paired gain over fixed B = 5,000 N·s/m; 9 states × 4 waves.
300 s per wave; known future waves; combined descriptive cohort
(original two + fresh two waves per state).

### Same plans. About half the time.

Direct evaluation vs POD reconstruction




Median; caps: 95th percentile · eight requests × ten repeats

8 fitting-state requests × 10 repeats; same initial POD screen.
Hs = 0.75 m, Tp = 3 s; Quadro RTX 5000, four lanes; warm timings.

### Explore the project

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**Paper, results & contact**

reuben-trade.github.io/wavex/

Earlier CEED paper and updated results are clearly identified on the project page.

Scope: known wave preview; uniform PTO damping; numerical validation. Next: causal forecasts, switched-plan validation and device testing. Industry mentors: Tim Green and Amal Jose Arackal.